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UNIVERSITY OF
WOLVERHAMPTON

6CS012 - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
HERALD COLLEGE
UNIVERSITY OF WOLVERHAMPTON

Tutorial - 01

An {almost} Complete Review of Matrix Operations and Derivative for Deep Learning.

Siman Giri {Module Leader - 6CS012}

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Instructions:

- This document is prepared to be standalone note on various concepts discussed in Lecture. Thus contains problem, exercise and background notes.
- You are expected to complete all the exercises, including example exercises if any.
- Example solutions are provided as a guide for solving the exercises. If you notice any typographical errors, bugs, or other issues, please inform your instructor and module leader.
{Your efforts will be hugely appreciated.}
- Exercises in Section 1: Preliminary Concepts will not be discussed in the tutorial sessions. Students are expected to complete these independently.

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1 Preliminary Concepts - Basic set Theory and Function

1.1 Understanding Set:

1. Definition of a Set

A **set** is a well-defined collection of unique or distinct objects.

Examples:

- A pack of wolves
- A deck of cards
- A flock of pigeons

2. Notation for Sets

- Sets are denoted by **capital letters** such as A or X .
- The elements (or members) of a set are listed using **curly brackets** $\{ \}$.
- Membership is denoted as:

$$x \in A \quad (\text{read: } x \text{ is an element of } A \text{ or } x \text{ belongs to } A)$$

3. Set Notations

There are several ways to define a set. The three most common notations are:

a. Verbal Notation

A set is described using words. Example:

Let X be the set of letters in the word *English*.

b. Roster Notation

A set is defined by explicitly listing its elements. Example:

$$X = \{x_1, x_2, \dots, x_n\}$$

c. Set-Builder Notation

A set is described using a property that its elements satisfy. Example:

$$X = \{x \mid x \text{ is an even integer and } 0 \leq x \leq 6\}$$

Equivalent to: $X = \{0, 2, 4, 6\}$

4. Relations Between Sets

a. Subset

A set A is a **subset** of B if every element of A is also an element of B . This means that A is contained within B .

Notation: $A \subseteq B$

Example: $\{4, 5, 8\} \subseteq \{2, 3, 4, 5, 6, 7, 8, 9\}$

b. Equal Sets

Two sets A and B are equal if they have exactly the same elements.

Notation: $A = B$

c. Empty Set

A set with no elements is called an **empty set** and is denoted by $\emptyset$.

d. Universal Set

The **universal set** (denoted by U) contains all elements under consideration, without any repetition.

5. Set Operations

a. Intersection

The **intersection** of sets A and B is the set of all elements common to both.

Notation: $A \cap B$

Definition: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

b. Union

The **union** of sets A and B is the set of all elements in either A or B , or both.

Notation: $A \cup B$

Definition: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

c. Disjoint Sets

Two sets A and B are **disjoint** if they have no elements in common.

Condition: $A \cap B = \emptyset$

d. Complement of a Set

The **complement** of set A , denoted A' , consists of all elements in the universal set U that are not in A .

Definition: $A' = \{x \mid x \in U \text{ and } x \notin A\}$

e. Difference of Two Sets

The **difference** of sets A and B , denoted $A \setminus B$, consists of elements that are in A but not in B .

Definition: $A \setminus B = A \cap B' = \{x \mid x \in A \text{ and } x \notin B\}$

6. Properties of Set Operations

- **Commutative Property:** $A \cup B = B \cup A$ and $A \cap B = B \cap A$
- **Associative Property:** $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$
- **Distributive Property:** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

7. Venn Diagrams

Venn diagrams are used to visually represent the relationships between sets. Below is an example of a Venn diagram showing the intersection and union of two sets:

1.2 Understanding Cartesian Product and Functions:

1. Cartesian Product of Sets

Given two sets A and B , we can define a new set called the **Cartesian product** of A and B , denoted as $A \times B$, which is the set of all ordered pairs where the first element belongs to A and the second element belongs to B :

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Example:

If $A = \{x, y\}$, $B = \{1, 2, 3\}$, and $C = \emptyset$, then

$A \times B = \{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}$.

and,

$A \times C = \emptyset$

Figure 1: Example - Cartesian Product.

Cartesian Product of Multiple Sets

The Cartesian product can be extended to multiple sets. The Cartesian product of n sets $A_1, A_2, \dots, A_n$ is defined as:

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

2. Mappings, Relations, and Functions

Mappings, also known as **relations** or **functions**, are subsets of Cartesian products. Given the Cartesian product $A \times B$, a function f is defined as:

$$f \subset A \times B.$$

This represents a special type of relation where $(a, b) \in f$ if, for every element $a \in A$, there exists a unique element $b \in B$. In other words, a function assigns a unique output in B for every input in A .

Function Notation

A function is denoted as:

$$f : A \rightarrow B$$

For ordered pairs $(a, b) \in A \times B$, the function notation is:

$$f(a) = b \quad \text{or} \quad f : a \rightarrow b.$$

Domain and Range

- The set A is called the **domain** of f . - The **range** (or **co-domain**) of f is given by:

$$f(A) = \{f(a) \mid a \in A\} \subset B.$$

The elements of the function's domain are considered **input values**, and the elements in the function's range are the corresponding **output values** also called the **image under**.

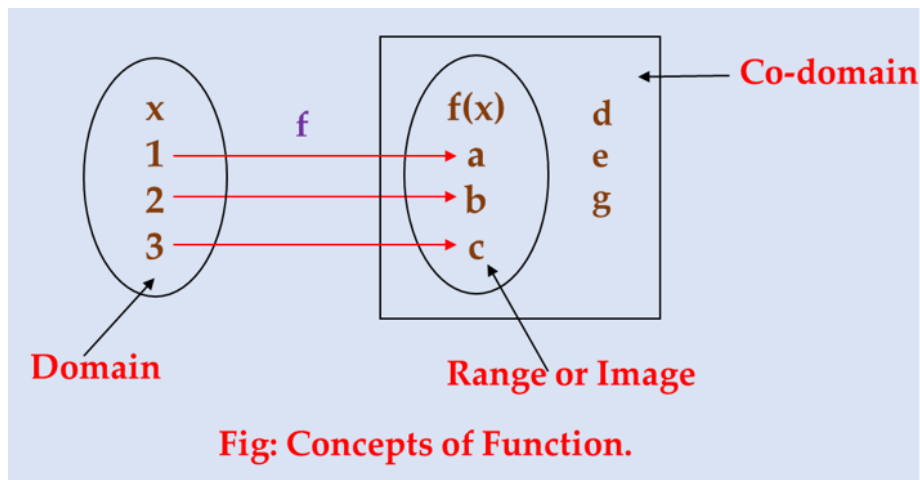


Figure 2: Concepts of Function.

3. A Valid Functions:

Let $f : A \rightarrow B$ be a **function** from the domain A to the codomain B . A function is said to be **well-defined** if each element in the domain is assigned to a unique element in the range. That is, a function must always return the same value for a specific input (**one-to-one**) or a collection of inputs (**many-to-one**).

Cases Where a Relation is Not a Function

A relation is **not** a function in the following cases:

1. There exists an element in the domain that has no associated image in the codomain.
2. There exists an element in the domain that has more than one associated image in the codomain (**one-to-many** relation).

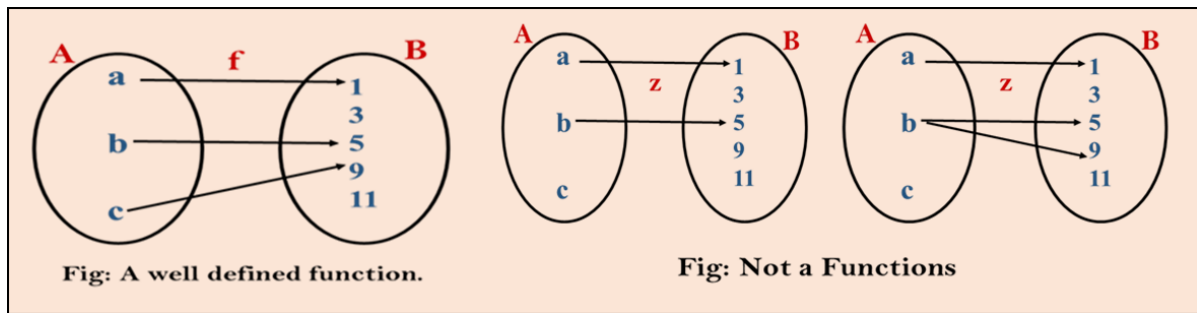


Figure 3: Example of Well defined and Invalid Function.

Composite Function:

A composite function is a function that is created by combining two functions. If you have two functions, say f and g , then the composite function $(f \circ g)(x)$ means you first apply function g to the input x and then apply function f to the result of $g(x)$. Mathematically:

$$(f \circ g)(x) = f(g(x))$$

In general, we have:

$$(f \circ g)(x) \neq (g \circ f)(x)$$

Example:

Suppose $\rightarrow f(x) = 2x + 3$ and $g(x) = x^2$

Then composite function would be:

$$(f \circ g)(x) = f(g(x)) = f(x^2) = 2x^2 + 3$$

Similarly:

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = (2x + 3)^2$$

Figure 4: Example Composite function.

Inverse Function:

The inverse of a function reverses the effect of the original function. If you have a function f , its inverse, denoted as f^{-1} , will "undo" the action of f . Mathematically, if:

$$y = f(x)$$

then the inverse function f^{-1} will satisfy:

$$x = f^{-1}(y)$$

Key Properties: If $f \circ f^{-1} = I$ and $f^{-1} \circ f = I$, this means that applying f and then f^{-1} or vice versa will give you the original input x , where I represents the identity function.

$$f^{-1}(x) \neq \frac{1}{f(x)}$$

For a function to have an inverse, it must be both one-to-one and onto. Essentially, every output of f must correspond to exactly one input.

Example of Inverse Function:

A good example of inverse functions are exponential and logarithmic functions: logarithms are the inverse of exponential functions and vice versa. We can say that:

- The domain of logarithmic functions is the range of exponential functions $(0, \infty)$,
- And the range of logarithmic functions is the domain of exponential functions $(-\infty, \infty)$.

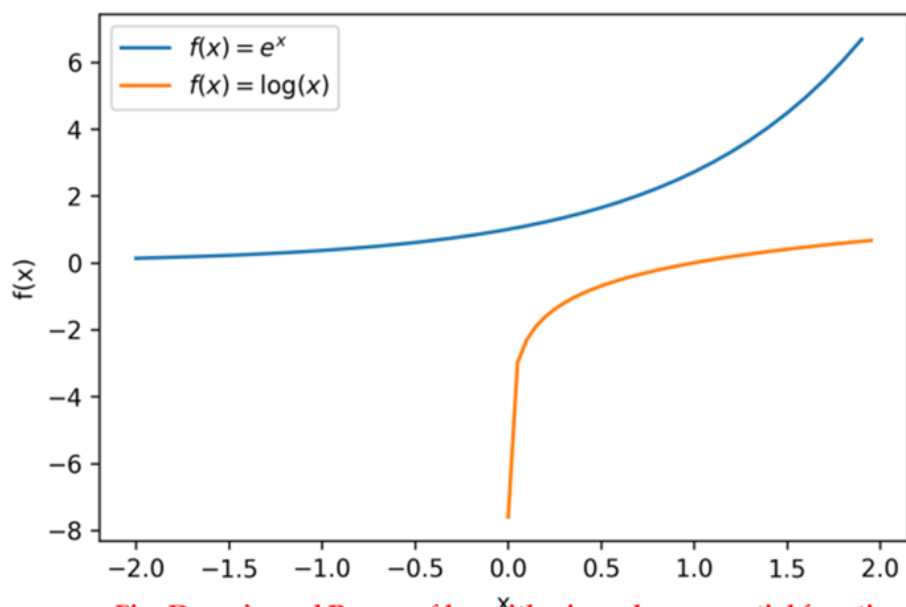


Fig: Domain and Range of logarithmic and exponential functions

Figure 5: Example of Inverse Function. For instance Consider the following exponential function:

$$f(x) = e^x$$

It's inverse is the logarithm defined as $f^{-1}(x) = \log(x)$

1.3 Motivation to Learn Math for Deep Learning.

The ultimate goal of **deep or machine learning** is **learning a function from data**, i.e., mappings from the domain **(feature vector space/set)** onto the range **(target variable)** of a function.

The objective of 6CS012 is to be able to understand all the highlighted terms in the above statement

1.4 Practice Problem on Preliminary Concepts:

1. Suppose that:

$$\begin{aligned} A &= \{x : x \in \mathbb{N} \text{ and } x \text{ is even}\}, \\ B &= \{x : x \in \mathbb{N} \text{ and } x \text{ is prime}\}, \text{ and} \\ C &= \{x : x \in \mathbb{N} \text{ and } x \text{ is a multiple of 5}\}. \end{aligned}$$

Describe Following Sets:

- (a) $A \cap B$ (The intersection of sets A and B):
- (b) $B \cap C$ (The intersection of sets B and C):
- (c) $A \cup B$ (The union of sets A and B):
- (d) $A \cap (B \cup C)$ (The intersection of set A and the union of sets B and C):

2. Prove:

$$\begin{aligned} i) A \cup (B \cap C) &= (A \cup B) \cap (A \cup C) \\ ii) A \cap (B \cup C) &= (A \cap B) \cup (A \cap C) \end{aligned}$$

Sample Proof for (i):

Proof of Set Equality

To prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, we need to show that both sides of the equation contain exactly the same elements. We can do this by proving two inclusions:

Inclusion One : $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$

Inclusion Two : $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$

Inclusion One - Proof

1. Proving $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$:

Take an arbitrary element $x \in A \cup (B \cap C)$. By definition of union, this means that $x \in A$ or $x \in (B \cap C)$.

- **Case 1:** If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$, because $A \subseteq A \cup B$ and $A \subseteq A \cup C$. Thus, $x \in (A \cup B) \cap (A \cup C)$.

- **Case 2:** If $x \in B \cap C$, then $x \in B$ and $x \in C$. Since $x \in B$, it follows that $x \in A \cup B$. Similarly, since $x \in C$, it follows that $x \in A \cup C$. Therefore, $x \in (A \cup B) \cap (A \cup C)$.

Since both cases imply that $x \in (A \cup B) \cap (A \cup C)$, we have shown that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

Inclusion Two - Proof

2. Proving $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$:

Take an arbitrary element $x \in (A \cup B) \cap (A \cup C)$. By definition of intersection, this means that $x \in A \cup B$ and $x \in A \cup C$.

- **Case 1:** If $x \in A$, then $x \in A \cup (B \cap C)$, because $A \subseteq A \cup (B \cap C)$.

- **Case 2:** If $x \notin A$, then since $x \in A \cup B$ and $x \notin A$, it must be that $x \in B$. Similarly, since $x \in A \cup C$ and $x \notin A$, it must be that $x \in C$. Therefore, $x \in B \cap C$, and so $x \in A \cup (B \cap C)$. Since both cases imply that $x \in A \cup (B \cap C)$, we have shown that $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

Conclusion

Since we have shown both inclusions, we conclude that:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Thus, the equality is proven.

3. Define a function: $f : \mathbb{N} \rightarrow \mathbb{N}$ a) that is one-to-one but not onto. b) that is onto but not one-to-one.
Solution:

Sample Solution - a

One such function is:

$$f(n) = 2n$$

where $f : \mathbb{N} \rightarrow \mathbb{N}$ is defined by multiplying each natural number by 2.

Explanation:

One-to-One (Injective):

A function is one-to-one if different inputs map to different outputs. Here, if $f(n_1) = f(n_2)$, then $2n_1 = 2n_2$, which implies $n_1 = n_2$. Therefore, $f(n) = 2n$ is injective.

Not Onto (Surjective):

A function is onto if every element in the codomain has a corresponding element in the domain. Here, the codomain is $\mathbb{N}$ (the set of all natural numbers). However, $f(n) = 2n$ only produces even numbers in $\mathbb{N}$, so there is no $n \in \mathbb{N}$ that maps to any odd natural number. Therefore, the function $f(n) = 2n$ is not onto.

Conclusion:

The function $f(n) = 2n$ is one-to-one but not onto.

2 Matrix Operation and System of Linear Equations.

2.1 Practice Problem on Basic Matrix Operation:

- The diagonal of a matrix A are the entries a_{ij} ; where $i = j$.
 - Write down the three-by-three matrix with ones on diagonal and zeros else where.
 - Write down the three-by-four matrix with ones on diagonal and zeros elsewhere.
 - Write down the four-by-three matrix with ones on the diagonal and zeros elsewhere.

Solution:

Sample Solution

$$a = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \square$$

- Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ Compute AD and DA .{observe and Explain the Results}.
- Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$. Verify that $AB = AC$.
- Prove that: $(AB)^T = B^T A^T$.

Review some Definitions:

- Let A be an m -by- n matrix with elements a_{ij} and b be an n -by- p matrix with elements b_{ij} , Then $C = AB$ is an m -by- p matrix and its ij elements $c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$.

Notice: second index of a and the first index of b are summed over.

$$\blacksquare A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \text{ then } A^T = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{bmatrix}. \text{ In other words: } a_{ij}^T = a_{ji}$$

- Let A be a rectangular matrix given by $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$. Compute $A^T A$ and show that it is a symmetric square matrix. Observe that the sum of the diagonal elements of $A^T A$ is the sum of squares of the elements of A .

6. Find the inverse of the matrices $A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}; B = \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix}$.

Inverse can be written as:

- We know:

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\det(A)} \cdot \begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{21} \end{bmatrix} = \frac{1}{ad - bc} \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

7. Given matrices A and B below, verify that they are inverses.

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}; B = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$$

Solution:

Hint:

- The matrices are inverses if the product AB and BA both equal the identity matrix $I_{2 \times 2}$.

8. Verify that given matrices are inverses of each other:

$$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix}; \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

3 System of Linear Equations

A **system of linear equations** (or **linear system**) is a collection of one or more linear equations involving the same variables. For example:

$$\begin{cases} -x_1 + 3x_2 = 7, \\ 3x_1 + 2x_2 = 1, \end{cases} \quad (1)$$

is a system of two equations with two unknown variables x_1, x_2 .

To solve a system of linear equations means to find such values of the variables x_1, x_2 , that all of its equations are simultaneously satisfied. There are various ways to solve system of linear equation mostly:

1. With Graph.
2. With Matrix.

A linear system is **inconsistent** if it has **no solution**, and otherwise it is said to be **consistent**. Consistent system can have **one or infinite number** of solutions.

3.1 Representation: System of Linear Equation

The first step in solving any system of Linear Equation is to write the problems in standard equation, present them in system of equation format and represent them in the form of matrix, further then various matrix methods like elimination or inverse could be used to solve such system of linear equation.

Following are some of the problem for all:

- write standard equation
- present them in the form of system of linear equation
- represent them in the form of matrix.

Example 1:

You have three types of fruits: apples, bananas, and cherries. You distribute them into three baskets as follows:

1. In Basket 1, you place 2 apples, 3 bananas, and 1 cherry, which weigh a total of 12 grams.
2. In Basket 2, you include 1 apple, 2 bananas, and 4 cherries, with a total weight of 10 grams.
3. In Basket 3, you add 3 apples, 1 banana, and 2 cherries, weighing a total of 14 grams.

Let:

- a be the weight of one apple (in grams.)
- b be the weight of one banana (in grams.)
- c be the weight of one cherry (in grams.)

Tasks:

- Set up a system of equations based on the information provided.
- Represent the system of equations in Matrix form.

Solution:

- **Step 1: Set Up the Equations:** From the information provided, we can write the following equations based on the contents and total weights of each basket:

1. Basket 1:

$$2a + 3b + 1c = 12 \quad (1)$$

2. Basket 2:

$$1a + 2b + 4c = 10 \quad (2)$$

3. Basket 3:

$$3a + 1b + 2c = 14 \quad (3)$$

■ Step 2: Write the System of Equations:

$$2a + 3b + c = 12$$

$$a + 2b + 4c = 10$$

$$3a + b + 2c = 14 \quad (1)$$

■ Step 3: Present in the Matrix Form:

$$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 4 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 12 \\ 10 \\ 14 \end{bmatrix}$$

3.2 Solving System of Linear Equation Using Matrices:

There exist a multiple methods on solving system of linear equations using Matrix, some of them are namely:

- Elimination Approach:
 1. Gauss - Elimination Method.
 2. Gauss - Jordan Method.
- Matrix Inversion Method.
- Cramer's Rule.
- LU Decomposition Method.
- Iterative or Numerical Approach for large matrices.

As you have already discussed in detailed about many of the methods in your computational mathematics class. In this session we will only focus on Matrix Inversion Method.

1. Matrix Inversion Method

The inverse of a matrix A is another matrix, denoted A^{-1} , which, when multiplied with A , results in the identity matrix I . In other words, for a square matrix A , the inverse A^{-1} satisfies:

$$A \cdot A^{-1} = A^{-1} \cdot A = I$$

where I is the identity matrix of the same dimension as A , containing 1s along its diagonal and 0s elsewhere.

Conditions for the Existence of an Inverse

- **Square Matrix:** Only square matrices (with the same number of rows and columns) can have an inverse.
- **Non-Singular (Determinant is Non-Zero):** A matrix must be non-singular, meaning its determinant is not zero. If $\det(A) = 0$, the matrix A does not have an inverse.

2. Finding the Inverse - Using Adjugate and Determinant:

For a matrix A , the inverse can be calculated as:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

where $\text{adj}(A)$ is the adjugate matrix, formed by the cofactors of A .

Example: For a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The inverse is:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

if $ad - bc \neq 0$.

About a Determinant.

The determinant is a scalar value calculated from a square matrix that provides essential information about the matrix's properties. It is denoted as $\det(A)$ or $|A|$ for a matrix A . Determinants are crucial in linear algebra for several reasons:

1. **Non - Singularity:** If the determinant of a matrix is non-zero, the matrix is invertible (non-singular); if it is zero, the matrix is singular (non-invertible).
2. **Area and Volume Interpretation:** For 2×2 and 3×3 matrices, the determinant represents the area or volume scaling factor of the linear transformation defined by the matrix.
3. **System of Linear Equations:** The determinant helps determine whether a system of linear equations has a unique solution, infinitely many solutions, or no solution.

For 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is calculated as:

$$\det(A) = ad - bc$$

This concept extends to larger matrices, where the calculation involves more complex processes using minors and cofactors.

3. Finding the Inverse - Using Elementary Row Operations:

To find the inverse of a matrix using elementary row operations, follow these steps:

Step - 1 - Form the Augmented Matrix:

1. Begin with the $n \times n$ matrix A for which you want to find the inverse.
2. Form an augmented matrix by appending the identity matrix I_n of the same size to the right of A , so the augmented matrix is written as $[A_n | I_n]$. For example, if A is 3×3 matrix write:

$$\left[\begin{array}{ccc|ccc} a_{11} & a_{12} & a_{13} & 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 1 \end{array} \right]$$

Step - 2 - Transform the Augmented Matrix to Row Echelon Form:

1. Use valid row operations to transform the left side of the augmented matrix (i.e. the matrix A) into the identity matrix I_n .

Valid Row Operations are:

- swapping rows
- scaling rows
- adding or subtracting multiples of rows

2. Each row operation you perform on A is also applied to the identity matrix on the right side, which gradually transforms it into the inverse A^{-1} , if one exists.

Step - 3 - Check for the Identity Matrix:

1. Continue applying row operations until the left side of the augmented matrix matches the identity matrix I_n . The right side of the augmented matrix should then transform into a new matrix, denoted as B .
2. If the left side of the augmented matrix successfully becomes I_n then B on the right side is the inverse matrix A^{-1} .

Step - 4 - Conform the existence of the Inverse:

1. If, after performing row operations, the left side of the augmented matrix does not result in the identity matrix I_n then A is not invertible (it has no inverse).
This situation arises when the matrix A is singular, meaning its determinant is zero. In such cases, the system of equations associated with A would not have a unique solution.
2. If the left side of the augmented matrix is transformed into I_n , the matrix on the right side is the inverse A^{-1} . Otherwise, the inverse does not exist.

3.2.1 Solving the System of Linear Equation with Inversion Method:

To solve a system of linear equations with three variables using the inverse method, follow these steps:

Step - 1 - Write the system in Matrix Form:

1. Represent the system of linear equations $AX = B$, where A is the coefficient matrix, X is the column matrix of variables, and B is the column matrix of constants.
2. For Example, for equations:

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

the matrix form is:

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

Step - 2 - Find the inverse of the Matrix:

1. Calculate the inverse of a matrix A if exists denote as A^{-1} .
2. Only systems with non-singular (invertible) matrices can be solved using this method, as the determinant of A must be non-zero.

Step - 3 - Multiply both sides by A^{-1} :

1. To isolate $\mathbf{X}$, multiply both sides of the equation by A^{-1} .

$$A^{-1}AX = A^{-1}B.$$

2. Since $A^{-1}A = I$ this simplifies to:

$$X = A^{-1}B$$

Step - 4 - Compute and Verify:

1. To compute $X = A^{-1}B$, perform the matrix multiplication $A^{-1}B$ to obtain the values of X , which corresponds to the solutions for x, y, z .
This yields the solution as a column matrix representing the ordered triple x, y, z
2. Verify the Solutions: Substitute the values of x, y, z back into the original equations to ensure that they satisfy all three equations.

3.3 Understanding the Solution of a System of Linear Equation: Graphically.

A system of linear equations can have the following types of solutions:

1. One unique solution.
2. Infinitely many possible solution.
3. No solution.

A linear equation in two variables (here x_1 and x_2 is represented geometrically by a line which points (x_1, x_2) make up the collection of solutions of the equation. This is called the graph of the linear equation. In case of the system of two equations there will be two lines corresponding to each of the equations. Solution will be the intersection point of these lines if exist, below we will observe example of each.

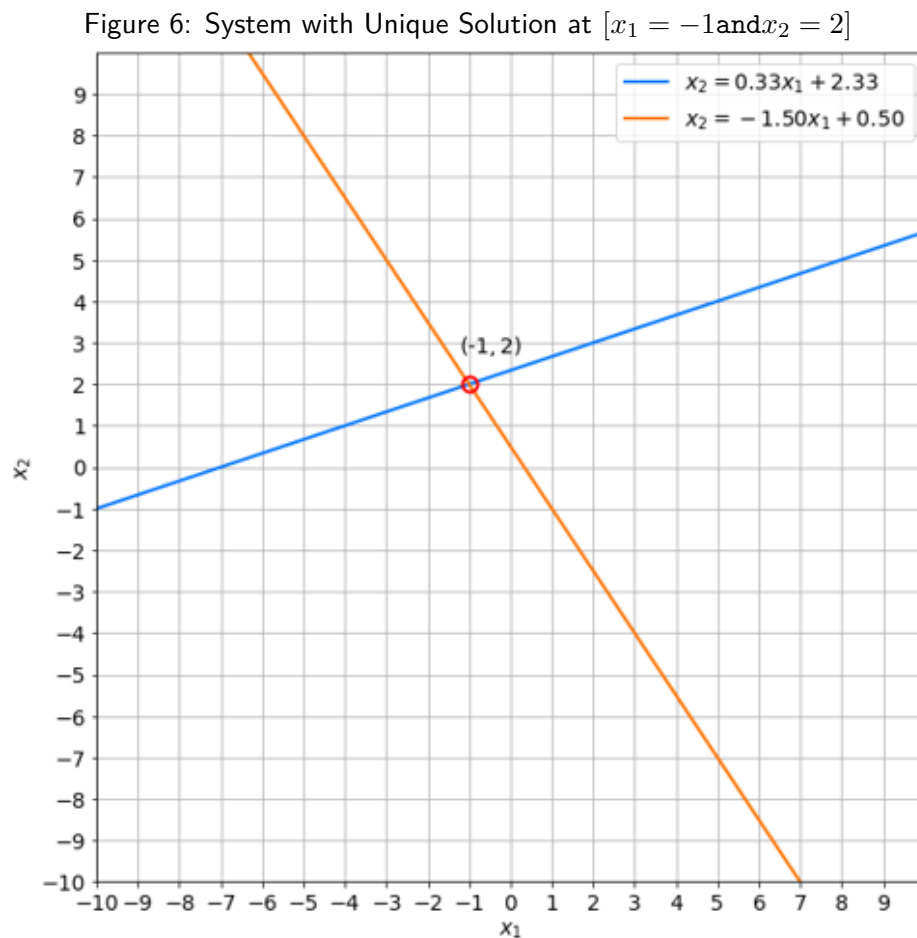
{ **Disclaimer!!!**The idea can be extended to three equations and three variables, in the case geometric representation will be **Plane**, and is not in the scope of this discussion}.

1. System with One Unique Solution:

For the following system of Linear Equation:

$$\begin{cases} -x_1 + 3x_2 = 7, \\ 3x_1 + 2x_2 = 1, \end{cases} \quad (1)$$

If we plot the above system of linear equation{range: -10 to 10 for both variables x_1 and x_2 , The plot will look like:



We can verify the above intersection by solving the system of equation using inverse method. For instance:

Solving with Matrix Inversion Method:

1. Step 1: Represent in the form $AX = B$:

$$\begin{bmatrix} -1 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

2. Step 2: Find A^{-1} using determinant and the formula is:

For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det(A) = ad - bc$$

Thus:

$$A^{-1} = \frac{1}{(-1)(2) - (3)(3)} \begin{bmatrix} 2 & -3 \\ -3 & -1 \end{bmatrix}$$

After further Simplification:

$$A^{-1} = \begin{bmatrix} -\frac{2}{11} & \frac{3}{11} \\ \frac{3}{11} & \frac{1}{11} \end{bmatrix}$$

3. Step 3: Solve $X = A^{-1}B$ i.e.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{2}{11} & \frac{3}{11} \\ \frac{3}{11} & \frac{1}{11} \end{bmatrix} \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

After further simplification we get:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Hence: Verified $\square$

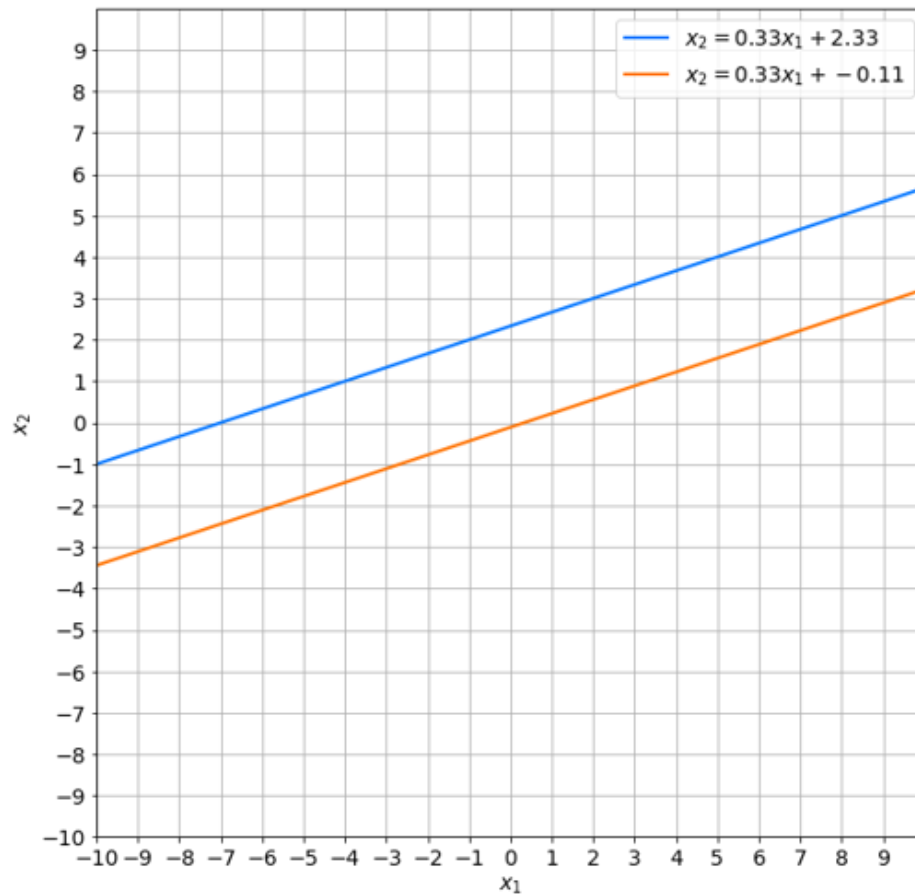
2. System with No Solution:

Plot the following system of Linear equation with range -10 to 10 for both the variables x_1 and x_2 :

$$\begin{cases} -x_1 + 3x_2 = 7 \\ 3x_1 - 9x_2 = 1 \end{cases} \quad (2)$$

A system of equation has no solution when the equations represent parallel lines that do not intersect and

Figure 7: System with No Solution



typically occurs under the following conditions:

- The equations are inconsistent, meaning they lead to contradiction for e.g. $0 = 5$.

Explanation - Why it has no Solution:

1. Step 1: Represent in the form $AX = B$:

$$\begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

2. Step 2: Find the A^{-1} :

- Estimating the determinant we get:

$$\det(A) = 0$$

this means the system either have infinite solutions or no solutions.

- If we continue to solve for A^{-1} using row reduction technique we get:

$$A^{-1} = \left[\begin{array}{cc|c} -1 & 3 & 7 \\ 3 & -9 & 1 \end{array} \right]$$

Applying $R_2 = R_2 + 3R_1$

$$\left[\begin{array}{cc|c} -1 & 3 & 7 \\ 0 & 0 & 22 \end{array} \right]$$

3. Step 3: Solve $X = A^{-1}B$ we get

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left[\begin{array}{cc|c} -1 & 3 & 7 \\ 0 & 0 & 22 \end{array} \right]$$

4. Step 4: Interpreting the Result:

The second row $0x_1 + 0x_2 = 22$ is a contradiction

i.e. no values for x_1 and x_2 can satisfy this equation.

Thus, the system is inconsistent, confirming that there is **no solution**.

□.

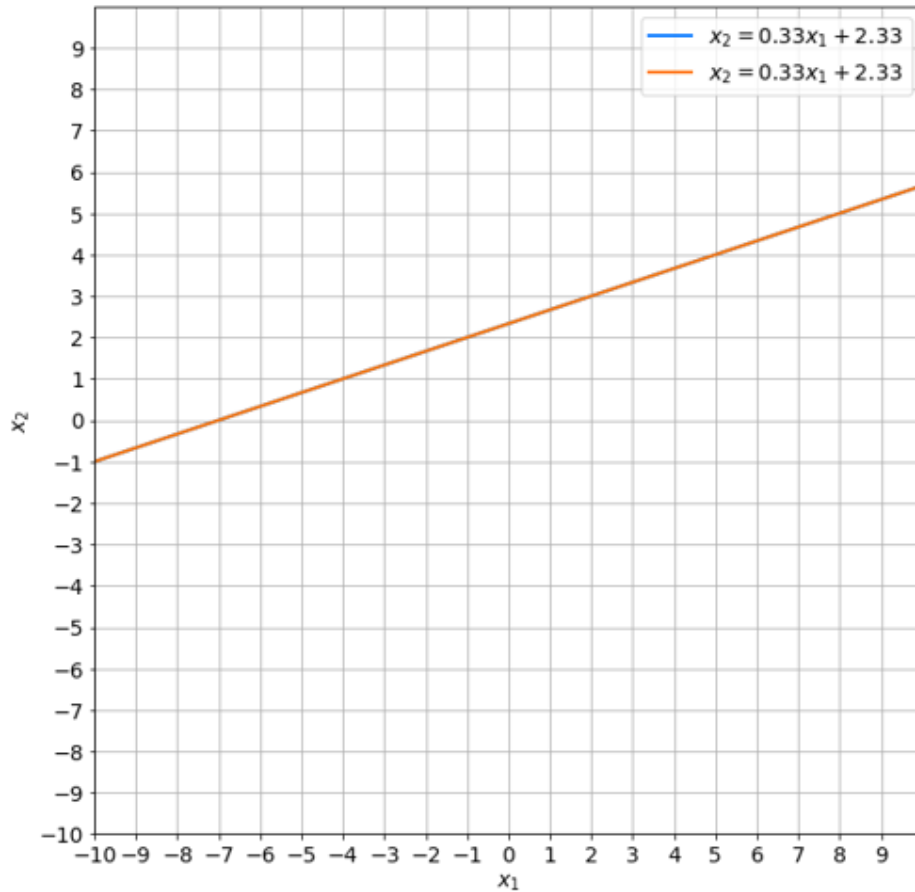
3. System with Infinitely Many Solutions:

Plot the following system of Linear equation with range -10 to 10 for both the variables x_1 and x_2 :

$$\begin{cases} -x_1 + 3x_2 = 7 \\ 3x_1 - 9x_2 = -21 \end{cases} \quad (2)$$

A system of equation has infinitely many solutions when the equations represent the same or when the planes

Figure 8: System with Infinitely Many Solutions



intersect in a line. This occurs under the following conditions:

- The equations are dependent, meaning that one equation can be expressed as a linear combination of the others. For example:

$$\begin{cases} x + y + z = 1 \\ 2x + 2y + 2z = 2 \end{cases} \quad (a)$$

The second equation in system of linear equation at {a} is multiple of the first, representing the same line.

Explanation - Confirming it has Infinitely Many Solutions:

1. Step 1: Represent in the form $AX = B$:

$$\begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ -21 \end{bmatrix}$$

2. Step 2: Applying a Row Reduction to verify: We can row-reduce the augmented matrix to determine if the system has a unique solution, no solution, or infinitely many solutions.

- The augmented matrix is:

$$\left[\begin{array}{cc|c} -1 & 3 & 7 \\ 3 & -9 & -21 \end{array} \right]$$

- Implementing Elementary Row Operations:

$$- R1 \rightarrow \frac{R1}{-1}$$

$$\left[\begin{array}{cc|c} -1 & -3 & -7 \\ 3 & -9 & -21 \end{array} \right]$$

$$- R2 \rightarrow R2 - \{-3 \times R1\}$$

$$\left[\begin{array}{cc|c} -1 & -3 & -7 \\ 0 & 0 & 0 \end{array} \right] \quad (1)$$

- Interpret the Result: The row - reduced form at {1} indicates that there is one free variable x_2 , which means system has infinitely many solutions.

3. Dependent Equation: The solution set in terms of free variable x_2 can be written as:

$$\begin{cases} x_1 = 3x_2 - 7 \\ x_2 = x_2 \end{cases} \quad (2)$$

So the general solution for the system is:

$$(x_1, x_2) = (3x_2 - 7, x_2)$$

Where x_2 is any real number. This confirms that the system has infinitely many solutions $\square$.

3.4 Practice Problem on Solving System of Linear Equation using Matrix Method:

1. System with Two Variables:

Express the system as $AX = B$, then solve using any methods of your choice.

{Our Recommendation: Use any of Matrix Method}

$$\text{a. } \begin{cases} x + 2y = 4 \\ 3x - 5y = 1 \end{cases}$$

{Answer: $x = 2$ and $y = 1$ }

$$\text{b. } \begin{cases} 5x + y = 13 \\ 3x + 2y = 5 \end{cases}$$

{ $x = 3$; $y = -2$ }

$$\text{c. } \begin{cases} 3x + 2y = -2 \\ x + 4y = 6 \end{cases}$$

{ $x = -2$; $y = 2$ }

$$\text{d. } \begin{cases} 2x + 3y = 6 \\ x - y = \frac{1}{2} \end{cases}$$

$x = \frac{5}{7}$ and $y = \frac{1}{7}$

$$\text{e. } \begin{cases} 4x + 3y = 11 \\ x - 3y = -1 \end{cases}$$

$x = 2$, and $y = 1$

$$\text{f. } \begin{cases} 2x + 3y = 8 \\ 4x + 6y = 20 \end{cases}$$

2. System with Three Variables:

$$\text{a. } \begin{cases} x + y - z = 2 \\ x + z = 7 \\ 2x + y + z = 13 \end{cases}$$

{ $x = 3$, $y = 3$, $z = 4$ }

$$\text{b. } \begin{cases} x + y - z = 2 \\ 3x + y = 7 \\ x + y + 2z = 3 \end{cases}$$

{ $x = 3$, $y = -2$, $z = 1$ }

$$\text{c. } \begin{cases} -x - 2y + z = 1 \\ 2x + 3y = 2 \\ y - 2z = 0 \end{cases}$$

{ $x = -1$, $y = 2$, $z = 1$ }

$$\text{d. } \begin{cases} x + 4y - z = 4 \\ 2x + 5y + 8z = 15 \\ x + 3y - 3z = 1 \end{cases}$$

{ $x = 3$, $y = 1$, $z = -1$ }

$$\text{e. } \begin{cases} 5x + 3y + 9z = -1 \\ -2x + 3y - z = -2 \\ -x - 4y + 5z = 1 \end{cases}$$

{ $x = -1$, $y = 2$, $z = 0$ }

$$\text{f. } \begin{cases} 6x - 5y + 2z = 3 \\ 2x + y - 4z = 5 \\ 3x - 3y + z = -1 \end{cases}$$

{ $x = 4$, $y = 5$, $z = 2$ }

Sample Solution:

Question:

$$\begin{cases} x + y - z = 2 \\ x + z = 7 \\ 2x + y + z = 13 \end{cases} \quad (\text{i})$$

Using Gauss Elimination Method:

1. Step 1: Write in the form $Ax = B$:

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 13 \end{bmatrix}$$

2. Step 2: Getting to Row Echelon form using Elementary row operation:

- Write in the Augmented Form:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 1 & 13 \end{array} \right]$$

- Performing Elementary Row Operation until we reached Row Echelon Form:

$$- R2 \rightarrow R2 - 1R1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 2 & 1 & 1 & 13 \end{array} \right]$$

$$- R3 \rightarrow R3 - 2R1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & -1 & 3 & 9 \end{array} \right]$$

$$- R3 \rightarrow R3 - 1R2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

- We have reached the Row Echelon Form, Thus rewriting the Matrix Form:

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} \quad (1)$$

3. Step 3: We can rewrite the matrix form at {1} as a linear combination of columns as:

$$x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} \quad (2)$$

4. Step 4: So the equation now can be written as:

$$\begin{cases} x + y - z = 2 \\ -y + 2z = 5 \\ z = 4 \end{cases} \quad (3)$$

5. Solving equation{3} By substitution we get:

$$\{x = 3, y = 3, z = 4\}.$$

Question:

$$\begin{cases} x - y + z = 6 \\ 2x + 3y = 1 \\ -2y + z = 5 \end{cases} \quad (\text{ii})$$

Using Matrix Inversion Method:

1. Step 1: Write in the form $Ax = B$:

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ 5 \end{bmatrix}$$

2. Step 2: Let's find A^{-1} using elementary row and augmented matrix technique:

▪ Write in the Augmented Form:

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \end{array} \right]$$

▪ Using Elementary Row operations to get to Reduced Row Echelon Form:

– $R2 \rightarrow -2 \times R1 + R2$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & 5 & -2 & -2 & 1 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \end{array} \right]$$

– $R2$ **SWAP** $R3$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \\ 0 & 5 & -2 & -2 & 1 & 0 \end{array} \right]$$

– $R2 \rightarrow \frac{R2}{-2}$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1/2 & 0 & 0 & -1/2 \\ 0 & 5 & -2 & -2 & 1 & 0 \end{array} \right]$$

– $R1 \rightarrow R1 + R2$ and $R3 \rightarrow -5 \times \{R2 + R3\}$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1/2 & 1 & 0 & -1/2 \\ 0 & 1 & -1/2 & 0 & 0 & -1/2 \\ 0 & 0 & 1/2 & -2 & 1 & 5/2 \end{array} \right]$$

– $R3 \rightarrow 2 \times R3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1/2 & 1 & 0 & -1/2 \\ 0 & 1 & -1/2 & 0 & 0 & -1/2 \\ 0 & 0 & 1 & -4 & 2 & 5 \end{array} \right]$$

$$- R2 \rightarrow \frac{1}{1} \times R3 + R2 \text{ and } R1 \rightarrow \frac{-1}{2} \times \{R3 + R1\}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -3 \\ 0 & 1 & 0 & -2 & 1 & 2 \\ 0 & 0 & 1 & -4 & 2 & 5 \end{array} \right]$$

- Thus the inverse of A is:

$$A^{-1} = \begin{bmatrix} 3 & -1 & -3 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix} \quad (1)$$

3. Step 3: Solving $AX = B$:

- Multiplying both sides of the Matrix by A^{-1} :

$$\begin{bmatrix} 3 & -1 & -3 \\ -2 & 1 & 2 \\ -4 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 & -1 & -3 \\ -2 & 1 & 2 \\ -4 & 2 & 5 \end{bmatrix} \begin{bmatrix} 6 \\ 1 \\ 5 \end{bmatrix} \quad (2)$$

- Solving By Multiplying the Matrices in equation{2}:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \quad (3)$$

- Solving equation {3} we get:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

4. Final Answer is:

$$x = 2, y = -1, \text{ and } z = 3 \quad \square.$$

3. Bonus Problem

1. Confirm why the following system has no solution:

$$\begin{cases} x + 2y - 3z = -1 \\ x - 3y + z = 1 \\ 2x - y - 3z = 2 \end{cases} \quad (a)$$

2. Confirm why the following system has no solution:

$$\begin{cases} 2x - 2y + 3z = 6 \\ 4x - 3y + 2z = 0 \\ -2x + 3y - 7z = 1 \end{cases} \quad (b)$$

3. Confirm why the following system has infinitely many solutions:

$$\begin{cases} x + 2y - z = 1 \\ 2x + 7y + 4z = 11 \\ x + 3y + z = 4 \end{cases} \quad (c)$$

4 Introduction to Derivatives in Machine Learning.

In machine learning, derivatives are fundamental in understanding how a function changes as its input changes, which is essential for tasks like optimization, gradient descent, and more. This tutorial will focus on reviewing key derivative concepts, covering scalar, vector, and matrix functions, and providing hands-on problems to illustrate how these concepts apply in machine learning.

{Disclaimer!!!: This tutorial focuses on behind the scene of optimization techniques required for machine learning. We will discuss Optimization in detail in future lectures when we discuss particular machine learning algorithms.}

What is Derivative?

The derivative of a function measures how the output value of the function changes as we make small adjustments to its input. In the simplest terms, if we have a function $f(x)$, the derivative f' at a point (x) tells us the rate of change of f at that point. This rate of change is crucial for optimization techniques, such as finding minima or maxima, which are frequently used in training machine learning models.

In this tutorial, we will review derivatives in the context of three types of functions:

- **Scalar Functions {Univariate}**: Function that take in and return scalar (single) values.
- **Multivariate Functions and Partial Derivative** Functions that take in scalars or vectors but output vectors, where gradients become useful.
- **Vector - Valued Function**: Functions that take in vectors and output matrices, where we often need the Jacobian matrix.

By the end, you'll have worked through examples to find critical points, understand gradients, and apply Jacobian in multi-dimensional scenarios. Let's start with the derivatives of scalar functions.

4.1 Derivatives of Scalar Functions:

A scalar function is a function that maps a real number x to another real number defined by a function $f(x)$. When we talk about the derivative of a scalar function, we are interested in the rate at which $f(x)$ changes as x changes.

Formal Definition of Derivative:

{Disclaimer!!! Even though we defined derivative under topics scalar derivatives, same is applied through out this discussion whether for Vector or Matrix valued Function}

Definition: First Rule of Derivative

For a function $f(x)$, the derivative at a point x , denoted as $f'(x)$ or $\frac{df}{dx}$, is defined by the limit {If the limit exists}:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This definition captures the **instantaneous rate of change** of $f(x)$ at x .

The derivative of a function at a point represents the slope of the tangent line at that point.i.e.

$$f'(a) = \text{"the slope of the tangent line to the graph of } f(x)$$

at $x = a$.

Derivatives as the Slope of a Function:

Imagine a curve representing a function $f(x)$ on a graph. At any specific point x on this curve, the slope of the tangent line to the curve at that point corresponds to the derivative $f'(x)$. This slope tells us how steeply the function is increasing or decreasing at that point.

Derivative can be used to understand the graph of any function such as $f(x)$:

- if $f'(x) > 0$, then the graph of $f(x)$ is increasing.
- if $f'(x) < 0$, then the graph of $f(x)$ is decreasing.
- if $f'(x) = 0$, then the graph of $f(x)$ has a horizontal tangent. This could indicate:
 1. A **local maximum** if $f(x)$ changes from increasing to decreasing i.e. $f'(x)$ changes from positive to negative around $x = c$.
 2. A **local minimum** if $f(x)$ changes from decreasing to increasing i.e. $f'(x)$ changes from negative to positive around $x = c$.
 3. A **point of inflection** if $f(x)$ does not change from increasing to decreasing or vice versa.
- **Cautions!!!!** $f'(x) = 0$ means there is horizontal tangent but this does not specify whether the point is a minimum, maximum or inflection without further information on $f'(x)$ or $f''(x)$ around $x = c$. $\{f''(x)$ is called second order derivative and is not scope of this course or this tutorial.}

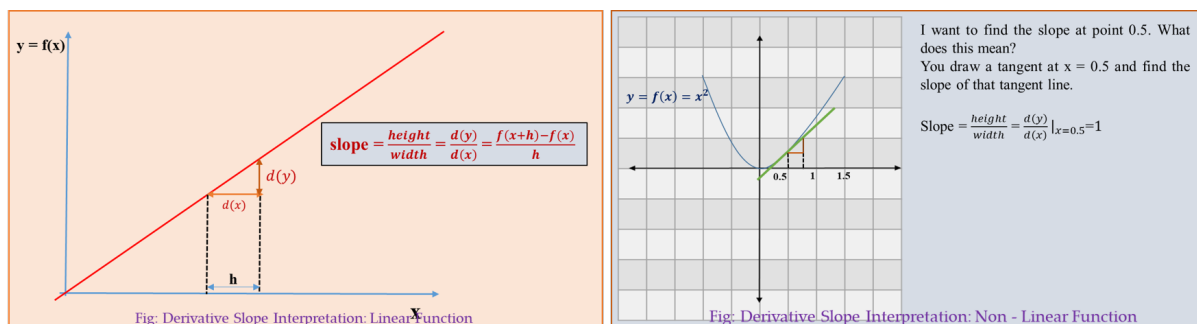


Figure 9: Slope Interpretation for Linear and Non Linear Function.

Some Rules of Determining Derivative:

{Disclaimer!!! Even though we defined rule under topics scalar derivatives, same is applied through out this discussion whether for Vector or Matrix valued Function}

Rule	Function	Derivative
Sum – Difference Rule	$f(x) \pm g(x)$	$f'(x) \pm g'(x)$
Multiplication by Constant	$c \cdot f(x)$	$c \cdot f'(x)$
Product Rule	$f(x) \cdot g(x)$	$f'(x) \cdot g(x) + f(x) \cdot g'(x)$
Quotient Rule	$\frac{f(x)}{g(x)}$	$\frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$
Chain Rule	$f(g(x))$	$f'(g(x)) \cdot g'(x)$

Figure 10: Rules for determining Derivative.

Known Derivative of some popular and common function:

Function - Type	Function - Notation	Derivative
Constant function	$f(x) = c$; where c is real constant.	$f'(x) = (c)' = 0$.
Identity function	$f(x) = x$	$f'(x) = (x)' = 1$.
Linear function	$f(x) = mx$	$f'(x) = (mx)' = m$.
Function of the form	$f(x) = x^n$	$f'(x) = (x^n)' = nx^{n-1}$.
Exponential function of the form	$f(x) = a^x$; where $a > 0$	$f'(x) = (a^x)' = a^x \ln(a)$.
Exponential function	$f(x) = e^x$	$f'(x) = (e^x)' = e^x$.
Logarithmic function	$f(x) = \ln(x)$	$f'(x) = (\ln(x))' = \frac{1}{x}$.
Sinusoidal function	$f(x) = \sin(x)$	$f'(x) = (\sin(x))' = \cos(x)$.
Cosine function	$f(x) = \cos(x)$	$f'(x) = (\cos(x))' = -\sin(x)$.
Tangent function	$f(x) = \tan(x)$	$f'(x) = (\tan(x))' = \sec^2(x)$.

Figure 11: Derivative of Some Common and Known Function.

4.1.1 Practice Problem: Let's Take Some Derivative:

Problem 1: Find $f'(x)$

1. $f(x) = x + \sqrt{x}$

Solution:

$$\begin{aligned} f'(x) &= \frac{d}{dx}\{x + \sqrt{x}\} \\ &= \frac{dx}{dx} + \frac{d\sqrt{x}}{dx} \\ &= 1 + \frac{d}{dx}\{x^{1/2}\} \\ &= 1 + (1/2)x^{(1/2)-1} \{\text{Power Rule}\} \\ &= 1 + (1/2)x^{-1/2} \\ &= 1 + \frac{1}{2\sqrt{x}} \quad \square. \end{aligned}$$

2. $f(x) = 2 + 1/x + 1/x^2$

3. $f(x) = \sqrt{x}\cos(x)$

4. $f(x) = x^3e^x$

5. $\frac{x^3}{e^x}$

6. $f(x) = (3x^2 + 2x + 1)^5$

How to apply a Chain Rule:

The chain rule is a fundamental technique in calculus used to find the derivative of a composite function—that is, a function within another function.

The chain Rule States that:

$$f'(x) = g'(u) \cdot u'(x)$$

Here:

$u(x)$: Inner Function

$g(u)$: Outer Function

- Summary of Steps Using the Chain Rule:
 - (a) Identify the Inner and Outer functions.
 - (b) Differentiate the outer function with respect to the inner function.
 - (c) Differentiate the inner function with respect to x .
 - (d) Apply the chain rule by multiplying the results from steps 2 and 3.
 - (e) Substitute back the inner function to get the final answer.

Solution:

Step - by - Step Solution - 6:

(a) Identify the inner and outer functions:

$$\text{Inner Function: } u = 3x^2 + 2x + 1$$

$$\text{Outer Function: } g(u) = u^5$$

(b) Differentiate the Outer function $g(u)$ w.r.t u :

$$g(u) = u^5$$

$$g'(u) = 5u^4 \quad \square \quad \{\text{Power Rule}\}$$

(c) Differentiate the Inner Function $u(x)$ w.r.t x :

$$u(x) = 3x^2 + 2x + 1$$

$$\begin{aligned} u'(x) &= \frac{d}{dx}(3x^2 + 2x + 1) \\ &= 6x + 2 \quad \square. \end{aligned}$$

(d) Apply the Chain Rule $f'(x) = g'(u) \cdot u'(x)$:

$$f'(x) = 5u^4 \cdot (6x + 2) \quad (1)$$

(e) Substitute $u = 3x^2 + 2x + 1$ back in equation{1}:

$$f'(x) = 5(3x^2 + 2x + 1)^4 \cdot (6x + 2) \quad \square. \quad \text{Final Answer.}$$

7. $f(x) = (\sin x)^4$

8. $f(x) = e^{(3x^2+2x)}$

9. $f(x) = \ln(5x^2 + 3x)$

10. $f(x) = \sqrt{x^3 + 4x + 1}$

Problem 2:

Suppose that $g(x) = x^2 - 3x + 2$ and $f(x)$ is a differentiable function. All we know about $f(x)$ is the following:

- $f(0) = 3$ and $f'(0) = -1$
- $f(1) = 5$ and $f'(1) = 0$
- $f(2) = -2$ and $f'(2) = 3$
- $f(3) = 6$ and $f'(3) = 1$

If possible:

1. Find the derivative of $f(x) \cdot g(x)$ at $x = 1$.
2. Find the derivative of $\frac{f(x)}{g(x)}$ at $x = 0$:
3. Find the derivative of $f(g(x))$ at $x = 0$:

Sample Solution

$$[f(g(x))]' = f'(g(x)) \cdot g'(x) \quad \{\text{Chain Rule}\}$$

At $x = 0$ and $g'(x) = 2x - 3$ and $g(0) = 0^2 - 3 \times 0 + 2 = 2$:

$$[f(g(x))]' = f'(g(0)) \cdot g'(0) = f'(2) \cdot [2 \cdot 0 - 3] = f'(2) \cdot (-3) = 3 \times -3 = -9 \quad \square$$

4.2 Application of Derivative:

Finding Maximum and Minimum Values Using Critical Points and Intervals:

The maximum and minimum values of a function within a specified interval refer to the highest and lowest points, respectively, that the function reaches on that interval. To determine these points, we often analyze the behavior of the function's derivative.

Here's how these terms apply based on the above context:

- **Maximum Value:** This is the greatest output value $f(x)$ that a function reaches within a specified interval. We determine this by evaluating the function at all critical points within the interval (where the derivative $f'(x)=0$ or is undefined) and at the interval endpoints, $x=a$ and $x=b$. The highest of these values is the maximum.
- **Minimum Value:** Similarly, the minimum is the smallest output $f(x)$ that the function reaches within the same interval $[a, b]$. By comparing the values at critical points and endpoints, the lowest result represents the minimum.

Steps to Find Maximum and Minimum Values:

- **Find the Derivative of the Function:** To determine where a function increases or decreases, first take its derivative. This will help identify the points where the function's slope is zero or undefined, which are potential points of interest.
- **Set the Derivative Equal to Zero:** Solve $f'(x) = 0$ to find the critical points. Critical points are where the function has a horizontal tangent line (slope = 0), indicating possible maximum, minimum, or saddle points.
- **Evaluate Critical Points within the Given Interval:** When an interval is specified (e.g., $[a, b]$), only consider the critical points within this range. If any critical points lie outside the interval, ignore them.
- **Check Function Values at Critical Points and Endpoints:**
 - Evaluate the original function $f(x)$ at each critical point within the interval.
 - Evaluate $f(x)$ at the endpoints of the interval, as maximum or minimum values could also occur at the boundaries.
- **Compare Values to Find Maximum and Minimum:** Compare all values obtained from the critical points and endpoints. The highest value will be the maximum, and the lowest value will be the minimum of the function within the specified interval.

About a critical points.

- Definition of Critical Points:

For a function $f(x)$:

- A critical point is any point $x = c$ in the domain of f where either:
 1. $f'(c) = 0$, or
 2. $f'(c)$ = does not exist or undefined.

- Why Critical Points Matter?

- Local Maxima and Minima: Critical points often help locate the highest and lowest points in a neighborhood, which is useful in optimization problems. For example, if $f(x)$ represents a profit model, the local maximum might represent the best profit conditions.
- Inflection Points: While not all critical points are inflection points, some critical points are places where the curve changes concavity (from concave up to concave down, or vice versa).

- How to Find Critical Points?

1. Take the derivative of the function, $f'(x)$.
2. Set the derivative equal to zero and solve for x to find where the slope is zero (horizontal tangent).
3. Check whether the points are inside defined interval.
4. If derivative is Undefined: Check at which point derivative is undefined, the points is also considered critical points.

4.2.1 Practice Problem on Maxima and Minima using derivative:

Problem 1: For the following function find the maximum or minimum of function at given interval using critical points, if exist:

1. $g(x) = x^2 - 4x + 5$ at interval $[1, 4]$ {Minimum at $g(2) = 1$ and Maximum at $g(4) = 5$.
2. $h(x) = \frac{x^2}{2} - 2x$ at interval $[-2, 3]$ {Minimum at $h(2) = -2$ and Maximum at $h(-2) = 6$ }
3. $p(x) = 3x^3 - 9x^2 + 7$ at interval $[-1, 2]$ {Minimum at $p(-1 \text{ or } 2) = -5$ and Maximum at $p(0) = 7$ }
4. $f(x) = x^2 - 4x + 6$ at interval $[0, 3]$ {Minimum at $f(2) = 2$ and Maximum at $f(0) = 6$ }
5. $q(x) = -2x^3 + 3x^2 + 12x$ at interval $[-1, 2]$ {Minimum at $q(-1) = -7$ and Maximum at $q(2) = 20$ }

Sample Solution for Finding Maximum and Minimum Points Using Critical Points:

Question:

$$f(x) = x^3 - 3x + 1 \quad \text{at interval } [0, 3]$$

Solution:

1. Find the derivative of function i.e. $f'(x)$:

$$f(x) = x^3 - 3x + 1$$

$$f'(x) = \frac{d}{dx}(x^3 - 3x + 1)$$

$$f'(x) = 3x^2 - 3$$

2. Find the critical points by setting $f'(x) = 0$:

$$f'(x) = 0$$

$$3x^2 - 3 = 0$$

Solving for x

$$3(x^2 - 1) = 0$$

$$(x - 1) \cdot (x + 1) = 0$$

So the critical points are $x = 1$ and $x = -1$ □.

3. Evaluate $f(x)$ at the interval $[0, 3]$:

Only the critical point $x = 1$ lies within this interval, thus we will ignore $x = -1$ because it is outside the interval.

$$f(1) = 1 - 3 \times 1 + 1 = -1$$

4. Evaluate $f(x)$ at the **Endpoints** i.e. at interval $[0, 3]$:

- $f(0) = 1$

- $f(3) = 19$

5. Therefore, the maximum value of $f(x)$ on the interval $[0, 3]$ is 19 at $x = 3$, and the minimum value is -1 at $x = 1$

4.3 Derivatives of Multivariate Functions.

Multivariate functions are functions that depend on two or more variables. For example, a function $f(x, y)$ that depends on two variables x and y can be analyzed using partial derivatives and directional derivatives. Understanding how these functions change with respect to their variables is essential in fields like optimization, economics, and machine learning.

Definition of Multivariate Function:

A multivariate function can be represented as:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

where n is the number of input variables. For example, a function of two variables can be written as:

$$f(x, y)$$

4.3.1 Partial Derivatives:

To determine the derivative of multivariate function we use partial derivative. Partial derivatives measure the rate of change of a multivariate function with respect to one variable while keeping all other variables constant. The **notation** for the partial derivative of f with respect to x is:

$$\frac{\partial f}{\partial x} \quad \text{or} \quad f_x$$

Example: Partial Derivative

Consider the function:

$$f(x, y) = x^2y + 3xy^2$$

- Partial derivative w.r.to x :

$$\frac{\partial f}{\partial x} = 2xy + 3y^2 \quad \{\text{Treating } y \text{ as a constant.}\}$$

- Partial derivative w.r.to y :

$$\frac{\partial f}{\partial y} = x^2 + 6xy \quad \{\text{Treating } x \text{ as a constant.}\}$$

Example from Lecture Slide:

Consider the function:

$$f(x, y) = x^2y$$

The partial derivative is determined as:

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial x} x^2 y = y \frac{\partial}{\partial x} x^2 = 2xy$$

Treat y as a constant, then take regular derivative.

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial y} x^2 y = x^2 \frac{\partial}{\partial y} y = x^2 \cdot 1$$

Treat x as a constant, then take regular derivative.

Derivative of $f(x, y) = x^2 y$ are $2xy; x^2$

Figure 12: Determining the Partial Derivative.

4.3.2 Gradient:

The gradient of a multivariate function is a vector that combines all the partial derivatives with respect to each variable. It represents the **direction of the steepest increase** in the function's value. The gradient, often denoted ∇f , is used to understand how the function changes in any direction within its domain.

For a function $f(x, y)$, the gradient is:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

This gradient vector points in the direction where f increases the **fastest**, and the **length (magnitude)** of the gradient gives the **rate of increase** in that direction.

Example of Gradient:

Find the Gradient of the function: $f(x, y) = x^2 + 3xy + y^2$:

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 2x + 3y \\ 3x + 2y \end{pmatrix}$$

If we want to know how $f(x, y)$ behaves at a specific point, say $(1, 2)$, we can substitute $x = 1$ and $y = 2$:

$$\nabla f(1, 2) = \begin{pmatrix} 2(1) + 3(2) \\ 3(1) + 2(2) \end{pmatrix} = \begin{pmatrix} 8 \\ 7 \end{pmatrix}$$

This gradient vector $\begin{pmatrix} 8 \\ 7 \end{pmatrix}$ points in the **direction** where the function $f(x, y)$ **increases most rapidly** from the point $(1, 2)$.

Key Difference between Gradient and Partial Derivative:

Aspect	Partial Derivative	Gradient
Scope	Focuses on the rate of change with respect to one specific variable	Captures the rates of change with respect to all variables simultaneously; provides the direction of steepest ascent
Form	A scalar (single value) for each variable	A vector that aggregates all partial derivatives
Interpretation	Indicates how the function changes when only one variable is varied	Gives a comprehensive view of the function's rate of change and direction in multidimensional space

Table 1: Comparison of Partial Derivative and Gradient

Applications:

1. **Gradient Descent:** In optimization, the gradient is used to find the direction in which a function decreases most rapidly. Gradient descent algorithms use this to minimize functions in machine learning.
2. **Directional Derivative:** The gradient is used to compute directional derivatives, which are derivatives of the function in the direction of an arbitrary vector. This is critical in physics and engineering for analyzing changes in any specified direction.

4.3.3 Practice Problem on Gradients:

1. Evaluate the Gradient of a function at given points:

1. $f(x, y) = x^2 + y^2$ at:

(a)(0, 0) {ans:(0, 0)} (b)(1, 3) {ans:(2, 6)} (c)(-1, -5) {ans:(-2, -10)}

2. $f(x, y, z) = x^3z - 2y^2x + 5z$ at:

(a)(1, 1, -4) {ans:(-14, -4, 6)} (b)(0, 1, 0) {ans:(-2, 0, 5)} (c)(-3, -2, 1) {ans:(19, -24, -22)}

3. $m(x, y) = \ln(x^2 + y^2 + 1)$ at:

(a)(0, 0) {ans:(0, 0)} (b)(1, -1) {ans:(2/3, -2/3)} (c)(-2, 3) {ans:(-2/7, 3/7)}

Sample Solution 3:

To evaluate the magnitude of gradient we first need to find the derivative of $f(x, y)$. Since it has two variables we need to find partial derivative against x and y
The gradient of:

$$\nabla m(x, y) = \left[\frac{\partial m(x, y)}{\partial x}, \frac{\partial m(x, y)}{\partial y} \right]$$

(a) Step 1: Calculate the Partial Derivatives:

▪ Partial Derivative w.r.to x) i.e. $\frac{\partial m}{\partial x}$:

i. Identify Inner and Outer Function:

inner function: $u = x^2 + y^2 + 1$

outer function: $g(u) = \ln(u)$

ii. Differentiate outer and inner function:

Differentiate the outer function w.r.to u :

$$g(u) = \ln(u)$$

$$\{\text{Derivative of Logarithm.}\} g'(u) = \frac{1}{u}$$

Differentiate the inner function w.r.to x :

$$u(x) = x^2 + y^2 + 1$$

$\{y^2 + 1 \text{ is treated as constant.}\}$

$$u'(x) = 2x$$

iii. Combine these results:

$$\begin{aligned} \frac{\partial m}{\partial x} &= \frac{1}{x^2 + y^2 + 1} \cdot 2x \\ &= \frac{2x}{x^2 + y^2 + 1} \quad \square. \end{aligned}$$

- Partial Derivative w.r.to (y) i.e. $\frac{\partial m}{\partial y}$:
 - i. Identify Inner and Outer Function:

$$\begin{aligned}\text{inner function: } u &= x^2 + y^2 + 1 \\ \text{outer function: } g(u) &= \ln(u)\end{aligned}$$

- ii. Differentiate outer and inner function:

Differentiate the outer function w.r.to u :

$$g(u) = \ln(u)$$

$$\{\text{Derivative of Logarithm.}\} g'(u) = \frac{1}{u}$$

Differentiate the inner function w.r.to y

$$u(y) = x^2 + y^2 + 1$$

$\{x^2 + 1 \text{ is treated as constant.}\}$

$$u'(y) = 2y$$

- iii. Combine these results:

$$\begin{aligned}\frac{\partial m}{\partial y} &= \frac{1}{x^2 + y^2 + 1} \cdot 2y \\ &= \frac{2y}{x^2 + y^2 + 1} \quad \square.\end{aligned}$$

- (b) Step 2: Gradient of $m(x, y)$:

$$\begin{aligned}\nabla m(x, y) &= \left[\frac{\partial m(x, y)}{\partial x}, \frac{\partial m(x, y)}{\partial y} \right] \\ \nabla m(x, y) &= \left[\frac{2x}{x^2 + y^2 + 1}, \frac{2y}{x^2 + y^2 + 1} \right] \quad \square.\end{aligned}$$

- (c) Step 3: Evaluate the Gradient at Specific Points:

- i. At $(0, 0)$:

$$\nabla m(0, 0) = \left[\frac{2 \cdot 0}{0^2 + 0^2 + 1}, \frac{2 \cdot 0}{0^2 + 0^2 + 1} \right] = (0, 0) \quad \square.$$

- ii. At $(1, -1)$:

$$\nabla m(1, -1) = \left[\frac{2 \cdot 1}{1^2 + (-1)^2 + 1}, \frac{2 \cdot (-1)}{1^2 + (-1)^2 + 1} \right] = \left(\frac{2}{3}, -\frac{2}{3} \right) \quad \square.$$

- iii. At $(-2, 3)$:

$$\nabla m(-2, 3) = \left[\frac{2 \cdot (-2)}{(-2)^2 + 3^2 + 1}, \frac{2 \cdot 3}{(-2)^2 + 3^2 + 1} \right] = \left(-\frac{2}{7}, \frac{3}{7} \right) \quad \square.$$

2. Problem: Using the Gradient to Evaluate Maxima and Minima of a Multivariate Function over a Specific Interval:

For a given functions, evaluate the maximum and minimum values over the specified interval:

1. $m(x, y) = -x^2 - y^2 + 4x + 6y$ interval $x : [0, 4]$ and $y : [0, 5]$
{ans: $Max : 13$ at $(2, 3)$ and $Min : 0$ at $(0, 0)$ and $(4, 0)$.}
2. $g(x, y) = xy + x^2 - 4y + 6$ interval $x : [-2, 2]$ and $y : [1, 3]$
{ans: $Max : 12$ at $(-2, 3)$ and $Min : -14$ at $(-2, 4)$.}
3. $p(x, y) = x^3 - 3xy + 4$ interval $x : [0, 2]$ and $y : [0, 3]$
{ans: $Max : 12$ at $(2, 0)$ and $Min : -6$ at $(2, 3)$.}
4. $h(x, y) = 2x^2 + 3y^2 - 12x - 18y + 20$ interval $x : [1, 5]$ and $y : [0, 4]$
{ans: $Max : 10$ at $(1, 0)$ and $Min : -25$ at $(3, 3)$.}
5. $f(x, y) = x^2 + y^2 - 4x - 6y + 13$ interval $x : [1, 3]$ and $y : [1, 4]$
{ans: $Max : 5$ at $(1, 1); (3, 1)$ and $Min : 0$ at $(2, 3)$.}

Sample Solution - 5:

1. Step 1: Find the Gradient:

The gradient is given by:

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

Calculating the partial derivatives:

$$\begin{aligned} \frac{\partial f}{\partial x} &= 2x - 4 \\ \frac{\partial f}{\partial y} &= 2y - 6 \end{aligned}$$

So the gradient is:

$$\nabla f(x, y) = \begin{pmatrix} 2x - 4 \\ 2y - 6 \end{pmatrix}$$

2. Step 2: Find Critical Points by setting Gradient equal to zero:

$$2x - 4 = 0 \Rightarrow x = 2$$

$$2y - 6 = 0 \Rightarrow y = 3$$

Thus the critical points is: $(2, 3)$

3. Step 3: Evaluate at critical points:

Now we need to check if the critical point $(2, 3)$ is within interval $[1, 3] \times [1, 4]$. Since $(2, 3)$ lies within in specified intervals, we evaluate the function:

$$f(2, 3) = 2^2 + 3^2 - 4 \cdot 2 - 6 \cdot 3 + 13 = 0$$

4. Step 4: Evaluate at Endpoints:

- For (1, 1):

$$f(1, 1) = 1^2 + 1^2 - 4 \cdot 1 - 6 \cdot 1 + 13 = 5$$

- For (1, 4):

$$f(1, 4) = 1^2 + 4^2 - 4 \cdot 1 - 6 \cdot 4 + 13 = 2$$

- For (3, 1):

$$f(3, 1) = 3^2 + 1^2 - 4 \cdot 3 - 6 \cdot 1 + 13 = 5$$

- For (3, 4):

$$f(3, 4) = 3^2 + 4^2 - 4 \cdot 3 - 6 \cdot 4 + 13 = 2$$

5. Step 5: Final Comparison:

- $f(1, 1) = 5$
- $f(1, 4) = 2$
- $f(3, 1) = 5$
- $f(3, 4) = 2$
- $f(2, 3) = 0$

Thus, $Max : 5$ at (1, 1) and (3, 1); $Min : 0$ at (2, 3) $\square$.

4.4 Vector - Valued Function:

Definition of a Vector - valued Function:

In vector-valued functions, we consider functions that take one or more variables as input and produce a vector as output. Unlike scalar-valued functions, where the output is a single value, vector-valued functions yield a vector that has multiple components.

A vector valued function can be represented as:

$$f(x) = \begin{pmatrix} f_1(x) \\ f_2(x) \\ \dots \\ f_n(x) \end{pmatrix}$$

Where $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ and each component $f_i(x)$ is a scalar function of x .

For example, a function that takes a scalar t , often representing time and maps it to a position vector in three-dimensional space could like this:

$$r(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}$$

Derivative of a Vector - Valued Function:

For a vector-valued function

$$f(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \dots \\ f_n(t) \end{pmatrix}$$

the derivative with respect to t) is the derivative of each components function. This derivative describes how the output vector changes with respect to t .

$$f'(t) = \begin{pmatrix} f'_1(t) \\ f'_2(t) \\ \dots \\ f'_n(t) \end{pmatrix}$$

Example:

Consider the vector - valued function:

$$f(t) = \begin{pmatrix} t^2 \\ \sin(t) \\ e^t \end{pmatrix}$$

The derivative $f'(t)$ would be:

$$f'(t) = \begin{pmatrix} \frac{d}{dt}(t^2) \\ \frac{d}{dt}(\sin(t)) \\ \frac{d}{dt}(e^t) \end{pmatrix} = \begin{pmatrix} 2t \\ \cos(t) \\ e^t \end{pmatrix}$$

Each component is differentiated individually, and the result is a new vector describing the rate of change of f at any point t .

Gradient of a Vector - Valued Function:

For functions that map from $\mathbb{R}^n \rightarrow \mathbb{R}^m$ we can think about the derivative in terms of a **Jacobian matrix**. Each element of this matrix is a partial derivative of an output component with respect to an input variable.

For a vector-valued function:

$$f(x) = \begin{pmatrix} f_1(x_1, x_2, \dots, x_n) \\ f_2(x_1, x_2, \dots, x_n) \\ \dots \\ f_m(x_1, x_2, \dots, x_n) \end{pmatrix}$$

, the **Jacobian matrix** J_f is defined as:

$$J_f = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

The Jacobian matrix contains all the partial derivatives of each output component with respect to each input variable, providing a full picture of how the vector-valued function f changes as each input variable changes.

Example problem on Jacobian:

Consider a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by:

$$f(x, y) = \begin{bmatrix} x^2 + y \\ x - y^2 \end{bmatrix}$$

Find the Jacobian:

Solution:

To find the Jacobian, we compute the partial derivatives of each component function:

1. Step 1: For $f_1(x, y) = x^2 + y$:

- $\frac{\partial f_1}{\partial x} = 2x$
- $\frac{\partial f_1}{\partial y} = 1$

2. Step 2: For $f_2(x, y) = x - y^2$:

- $\frac{\partial f_2}{\partial x} = 1$
- $\frac{\partial f_2}{\partial y} = -2y$

3. Step 3: The Jacobian is:

$$j_f(x, y) = \begin{bmatrix} 2x & 1 \\ 1 & -2y \end{bmatrix}$$

Final Interpretation:

The Jacobian matrix describes how small changes in the input vector x lead to changes in the output vector $f(x)$. The Jacobian can also be thought of as the best linear approximation of f near a point in $\mathbb{R}^n$.

In machine learning, vector-to-vector derivatives (and thus Jacobians) are frequently used in areas such as backpropagation in neural networks, transformations in deep learning layers, and optimization algorithms that involve multiple outputs and inputs.

————— Where we go from there is a choice I leave to you – Neo{The Matrix}. —————

THE END